

2026-08-24 Candidate focal new hires and their presence in USPTO

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Introduction to this report

Universe sample of candidate new hires $\xRightarrow{\text{Match with inventors}}$ Matched sample of candidate new hires

There are 3 parts in this report.

- Section 1 offers summary statistics for [the universe sample of candidate new hires](#).
 - This is the broadest sample because the only sample restriction I made is to keep scientists and engineers.
 - Even though our main purpose is to understand match rates (to the inventor database) across industries/occupations, I think it is necessary to first characterize the baseline distribution.
- Section 2 describes to what extent can the broadest sample be matched to [the inventor database](#).
 - It mainly characterizes heterogeneous match rates across industries, occupations, and countries.
- Section 3 offers summary statistics for [the matched sample of candidate new hires](#).
 - By comparing section 1 and 3, we can partially understand how samples change when we shift attention to only those inventors present in USPTO.

Outline

1 Summary statistics for the universe sample of candidate focal new hires

The universe sample of candidate focal new hires

Occupation distribution: Messy classifications

Industry distribution

Other results on the universe sample of candidate focal new hires

2 Match rates

3 Summary statistics for the matched sample of candidate focal new hires

The construction of the universe sample of candidate focal new hires

- ① Keep all employment spells in the following two 2-digit occupation groups.¹
 - 17: Architecture and Engineering occupations.
 - 19: Life, Physical, and Social Science Occupations.
- ② Further retain employment spells starting from January 2021 to December 2023.
- ③ Exclude employment spells with missing geography or job title information.
- ④ Exclude internship positions.
- ⑤ Retain one employment spell within each user-company cell ([the universe sample of candidate focal new hires](#)).

This is the broadest sample of the new hires. We will impose further restrictions on industries and occupations, guided by the set of summary statistics that I will report later.

¹The two occupation groups correspond to 125 distinct ONET occupations. However, there are only 52 matched ONET occupations in Revelio's sample.

Some basic numbers

- Number of candidate focal new hires (user-company level): 16,423,317
- Number of distinct users: 13,943,692
- Number of distinct companies: 2,081,233
- Number of distinct countries: 226
- Number of ONET occupations: 52
- Number of NAICS 6-digit industries: 1,014

All plots in this report are simple averages over these user-company observations (no weights used).

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Occupation distribution: Messy classifications

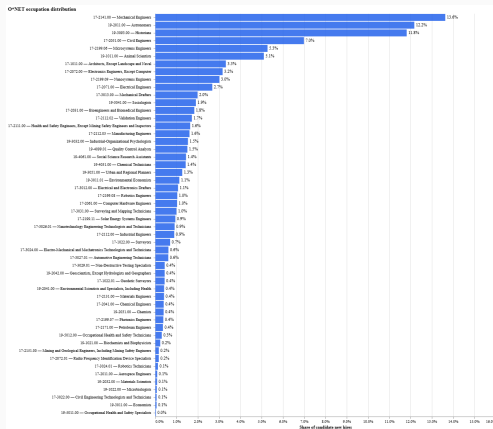
Industry distribution

Other results on the universe sample of candidate focal new hires

2 Match rates

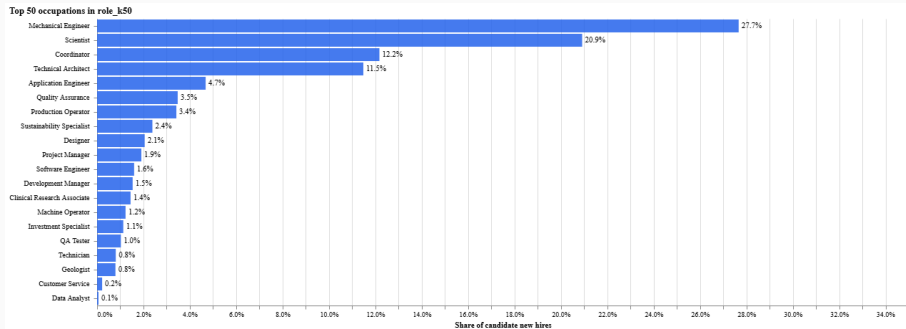
3 Summary statistics for the matched sample of candidate focal new hires

Occupation distribution in the sample of candidate new hires (ONET)



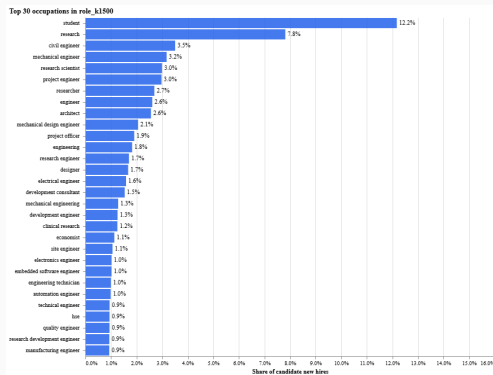
- Top 5 ONET occupations: Mechanical Engineers (13.6%), Astronomers (12.2%), Historians (11.8%), Civil Engineers (7.0%), Microsystems Engineers (5.3%).
- We have a very large share of “Astronomers” and “Historians”. This is related to the specific procedures how Revelio assigns the ONET occupation codes. I will talk more on this point later.

Occupation distribution (Revelio's own 50-category occupations)



- Top 5 Revelio's own occupations (50-category): Mechanical Engineers (27.7%), Scientists (20.9%), Coordinator (12.2%), Technical Architect (11.5%), Application Engineer (4.7%).
- They are more aligned with our expectations, but if we examine them carefully, they are still quite weird. For example, all 12.2% “Coordinators” in current Revelio's own occupation classification are classified as “Astronomers” in ONET.

Occupation distribution (Revelio's own 1500-category occupations)



- Top 5 Revelio's own occupations (1500-category): student (12.2%), research (7.8%), civil engineer (3.5%), mechanical engineer (3.2%), research scientist (3.0%).
- These 1500-category occupations are weird: we cannot judge the nature of the occupations simply based on their names.
- For example, the “student” role in Revelio's own classification is classified as “Astronomers” in ONET, the “research” role is classified as “Historians” in ONET – so the shares match.

How does Revelio assign an occupation to a user's employment spell?

How does Revelio assign an occupation to a user's employment spell?

- Revelio first construct the user's occupation (with different levels of aggregation²) based on their own occupation classification method (which is not public).
- Then they construct a crosswalk mapping from their own occupation classification (`role_k500`) to ONET occupations.
- The Revelio's way to construct occupations implies:
 - Each occupation in `role_k1500` is mapped to one and only one ONET occupation; while one ONET occupation could be mapped to multiple roles in `role_k1500`.
 - For the series of Revelio's own occupation classifications, a role in a finer classification (e.g., in `role_k500`) is mapped to 1 and only 1 role in a broader classification (e.g., in `role_k50`); while one role in a broader classification is mapped to multiple roles in a finer classification.

²Variables `role_k50`, `role_k150`, `role_k300`, `role_k500`, `role_k1000`, `role_k1500` represent Revelio's own occupation classifications with different levels of aggregation.

Messy occupation classifications

- Revelio's own occupation classifications are messy and often don't match to their literal meaning.
- Some unpleasant examples include:
 - student (in role_k1500) → Student Intern (in role_k150 to role_k1000) → Coordinator (in role_k50) → Astronomers (in ONET)
 - research (in role_k1500) → Researcher (in role_k1000) → Scientist (in role_k50 to role_k500) → Historians (in ONET)
 - research scientist (in role_k1500) → Scientist (in role_k50 to role_k1000) → Animal Scientists (in ONET)
 - thesis (in role_k1500) → Thesis Work (in role_k500 to role_k1000) → Scientist (in role_k50 to role_k300) → Animal Scientists (in ONET)
 - research technician (in role_k1500) → Scientist (in role_k50 to role_k1000) → Nanotechnology Engineering Technologists and Technicians (in ONET)

My suggestions on occupation classifications

- We shouldn't use a single Revelio's occupation variable for sample construction.
 - The ONET occupations are very unstable (recall the Astronomers and Historians).
 - We shouldn't either start with Revelio's own occupation classifications, because these variables often have ambiguous names (we cannot tell the exact nature of an occupation simply from its name itself).
- The best workflow is:
 - We start from a selected industry (or a set of industries) because Revelio's industry classifications seem to be more consistent across NAICS and their own industry classification system.
 - Next, using the results in Section 4 of this report, we select occupations that have relatively large share within the selected industry.
 - Optionally, we can do further restrictions based on users' reported job titles for robustness checks.
 - For example, exclude those new hires whose self-reported job titles are clearly not what we want.
 - For example, we keep only the job titles within the ONET codes that frequently show up in the sample.
- In summary, these are limitations inherited in Revelio's data, and we need to carefully deal with measurement errors of users' occupations.

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Occupation distribution: Messy classifications

Industry distribution

Other results on the universe sample of candidate focal new hires

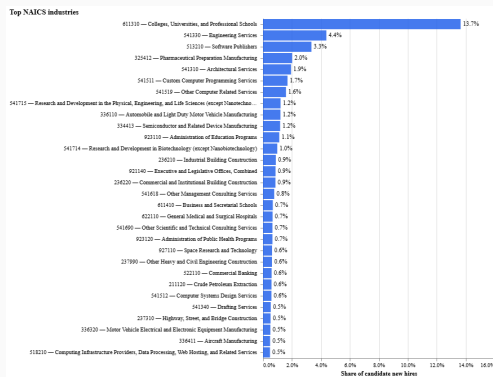
2 Match rates

3 Summary statistics for the matched sample of candidate focal new hires

The industry variables in the Revelio data

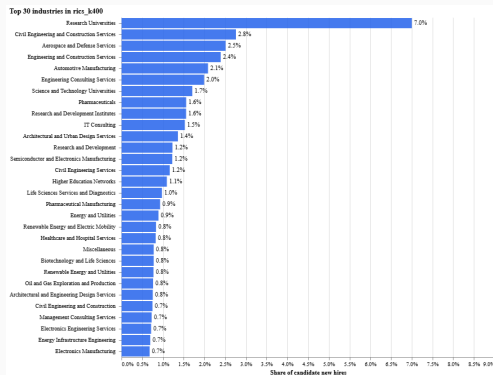
- There are 2 main sets of variables indicating a user's industry.
 - `naics`: The NAICS industry classifications at 6 digits (used mainly in North America: US, Canada).
 - `rics_k50`, `rics_k200`, `rics_k400`: Revelio's own industry classifications (50-, 200-, and 400-category).
- Revelio doesn't document how it constructs these industry variables.
- Unlike occupation classifications (where ONET codes are derived from its own occupation classifications), there is no definite relationship between NAICS industries and Revelio's own industries.
 - For example, "Pharmaceutical Preparation Manufacturing" (in NAICS) maps to multiple industries in `rics_k400`: Pharmaceuticals (54.5%); Pharmaceutical Manufacturing (32.7%); Biotechnology and Life Sciences (3.7%); Life Sciences and Diagnostics (2.5%); Biopharmaceuticals and Healthcare Services (1.8%).
 - In the other way around, "Pharmaceuticals" (in `rics_k400`) maps to multiple NAICS industries: Pharmaceutical Preparation Manufacturing (70.5%); Biological Product (except Diagnostics) Manufacturing (13.7%); Research and Development in Biotechnology (except Nanobiotechnology) (6.1%); Surgical and Medical Instrument Manufacturing (5.3%).

Industry distribution (6-digit NAICS classifications)



- Top 5 NAICS industries: Colleges, Universities, and Professional Schools (13.7%); Engineering Services (4.4%); Software Publishers (3.3%); Pharmaceutical Preparation Manufacturing (2.0%); Architectural Services (1.9%).
- There are in total 1,014 6-digit NAICS industries in the sample.

Industry distribution (Revelio's own 400-category industry classifications)



- Top 5 Revelio's own industries (400-category): Research Universities (7.0%); Civil Engineering and Construction Services (2.8%); Aerospace and Defense Services (2.5%); Engineering and Construction Services (2.4%); Automotive Manufacturing (2.1%).

My suggestions on industry classifications

- Revelio's own industry classifications have meaningful names (unlike their own occupation classifications), so I suggest to start with these variables.³
- In future robustness checks, we can further restrict samples based on NAICS industries.

³We don't know the specific ways Revelio assigns an industry to a company or a user, so from this point of view, NAICS and Revelio's own industry classifications are the same. However, NAICS industries are much finer (which could indicate more potential measurement errors), so I prefer to use Revelio's own industry classifications.

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Occupation distribution: Messy classifications

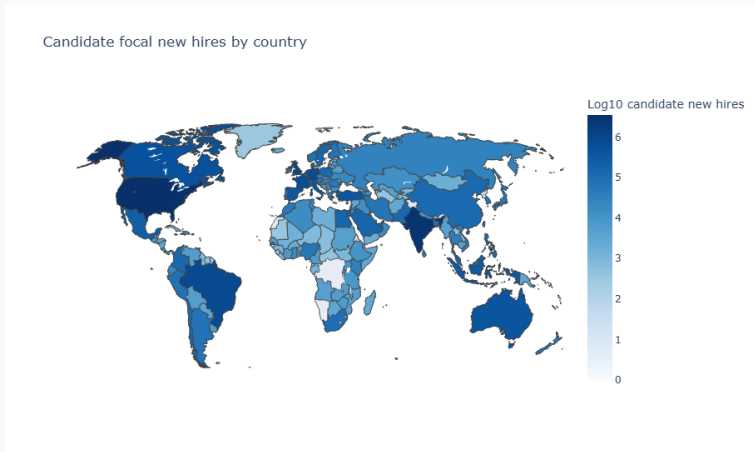
Industry distribution

Other results on the universe sample of candidate focal new hires

2 Match rates

3 Summary statistics for the matched sample of candidate focal new hires

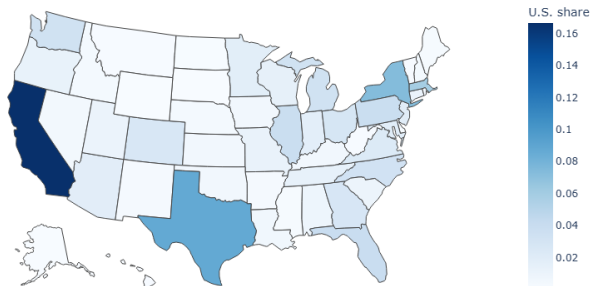
Geography distribution: Across countries



- Top 5 countries with most focal new hires: US (21.3%); India (16.1%); UK (6.0%); Brazil (4.9%); France (4.4%).

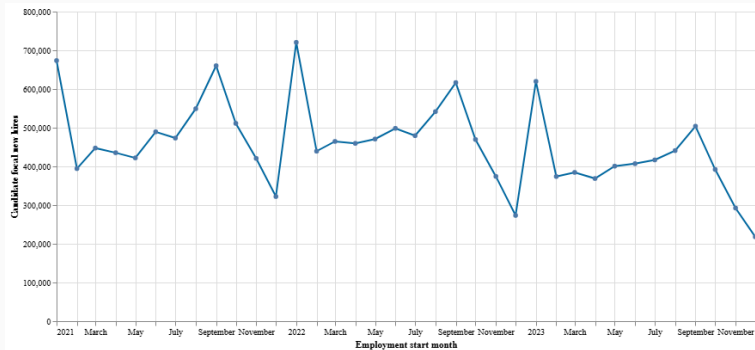
Geography distribution: Within US across states

Candidate focal new hires by U.S. state

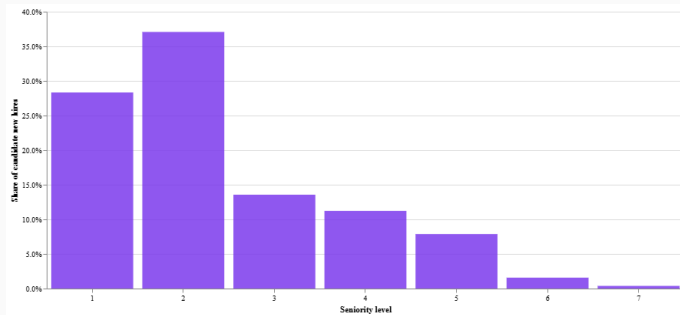


- Top 5 states with most focal new hires: California (16.7%); Texas (8.8%); New York (7.4%); Massachusetts (6.0%); Florida (4.0%).

Time series for the number of new hires (start time)



Seniority distribution



- The seven ordinal seniority levels are:
 - 1: Entry Level (e.g., Accounting Intern, Software Engineer Trainee, Paralegal)
 - 2: Junior Level (e.g., Account Receivable Bookkeeper, Junior Software QA Engineer, Legal Adviser)
 - 3: Associate Level (e.g., Senior Tax Accountant; Lead Electrical Engineer; Attorney)
 - 4: Manager Level (e.g., Account Manager; Superintendent Engineer; Lead Lawyer)
 - 5: Director Level (e.g., Chief of Accountants; VP Network Engineering; Head of Legal)
 - 6: Executive Level (e.g., Managing Director, Treasury; Director of Engineering, Backend Systems; Attorney, Partner)
 - 7: Senior Executive Level (e.g., CFO; COO; CEO)

Outline

① Summary statistics for the universe sample of candidate focal new hires

② Match rates

The matched sample of candidate focal new hires

Occupation distribution of match rates

Industry distribution

Other results on the matched sample

③ Summary statistics for the matched sample of candidate focal new hires

The inventor database

- Gaurav sent me a list of users that can be matched to an inventor ID (or multiple inventor IDs).
 - It is at user-level (if we temporarily ignore the small set of users who have multiple inventor IDs).
 - I will call it the [inventor database](#).
 - I merge the universe sample of candidate focal new hires with the inventor database.
- This section is mainly about heterogeneous match rates across occupations, industries, and countries.
 - Among the 16,423,317 candidate focal new hires (user-company level), there are 232,096 matched observations (match rate = 1.4%).
 - Among the 13,943,692 distinct users in the sample of candidate focal new hires (user level), there are 197,909 matched users (match rate = 1.4%).
 - The match rate for US at the user-company level is 3.9% (at user level is 3.8%).
 - The match rate for non-US at the user-company level is 0.7% (at user level is 0.8%).
- In what follows, I will report match rates at the user-company level.

Outline

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② Match rates

The matched sample of candidate focal new hires

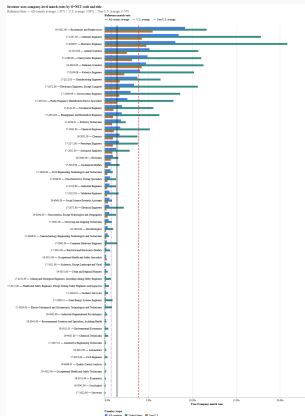
Occupation distribution of match rates

Industry distribution

Other results on the matched sample

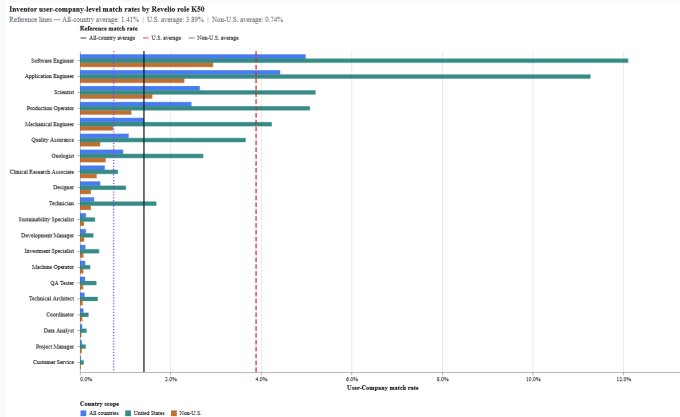
③ Summary statistics for the matched sample of candidate focal new hires

ONET occupation distribution of match rates



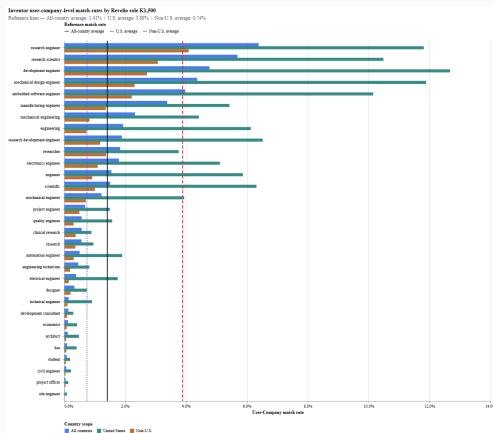
- Top 5 ONET occupations with the highest match rates: Biochemists and Biophysicists (9.18%); Materials Engineers (8.45%); Photonics Engineers (8.07%); Animal Scientists (5.13%); Nanosystems Engineers (4.90%).
- Match rates are higher within US new hires for 51 of the 52 ONET occupations. Occupational Health and Safety Specialists are the sole exception.

Revelio's own 50-category occupation distribution of match rates



- Top 5 Revelio's own 50-category occupations with the highest match rates: Software Engineer (4.98%); Application Engineer (4.42%); Scientist (2.65%); Production Operator (2.46%); Mechanical Engineer (1.42%).
- Again, match rates are higher within US new hires.

Revelio's own 1500-category occupation distribution of match rates



- Top 5 Revelio's own 1500-category occupations with the highest match rates: research engineer (6.39%); research scientist (5.69%); development engineer (4.77%); mechanical design engineer (4.37%); embedded software engineer (4.37%).

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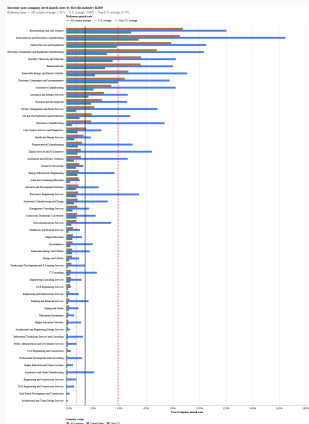
Occupation distribution of match rates

Industry distribution

Other results on the matched sample

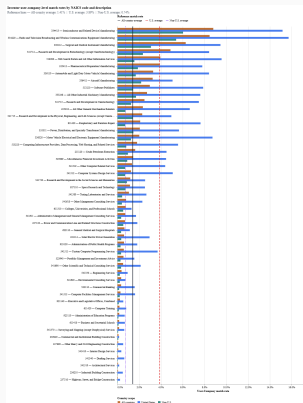
③ Summary statistics for the matched sample of candidate focal new hires

Revelio's own 400-category industry distribution of match rates



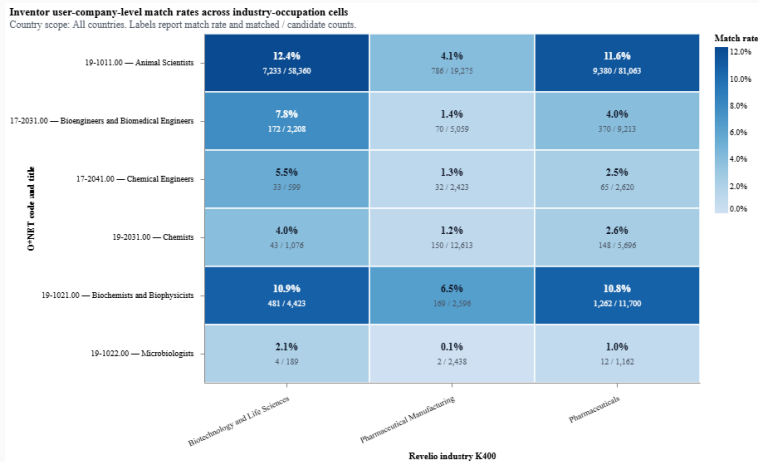
- Top 5 displayed Revelio 400-category industries with the highest match rates: Biotechnology and Life Sciences (8.77%); Semiconductor and Electronics Manufacturing (8.54%); Medical Devices and Equipment (7.89%); Electronic Components and Equipment Manufacturing (6.80%); Specialty Chemicals and Materials (5.61%).

NAICS (6 digits) industry distribution of match rates



- Top 5 NAICS (6 digits) industries with the highest match rates: Semiconductor and Related Device Manufacturing (8.81%); Radio and Television Broadcasting and Wireless Communications Equipment Manufacturing (8.47%); Surgical and Medical Instrument Manufacturing (6.27%); Research and Development in Biotechnology (except Nanobiotechnology) (4.88%); Web Search Portals and All Other Information Services (3.95%)

Match rates in our baseline BioPharm industry



- X-axis is Revelio's own industry classification, and y-axis is ONET occupations. I present the number of candidate focal new hires, the number of matched new hires, and the match rates in each industry-occupation cell.

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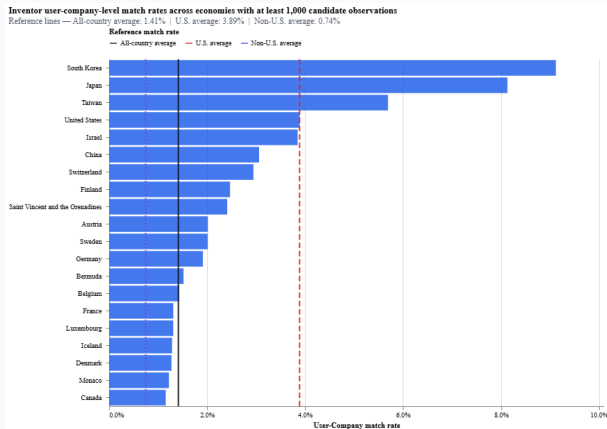
Occupation distribution of match rates

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Other results on the matched sample

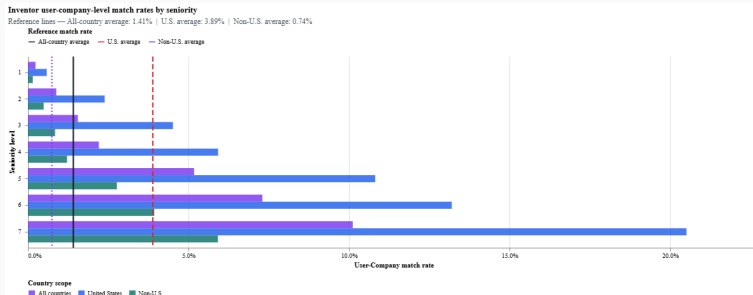
③ Summary statistics for the matched sample of candidate focal new hires

Match rates across countries



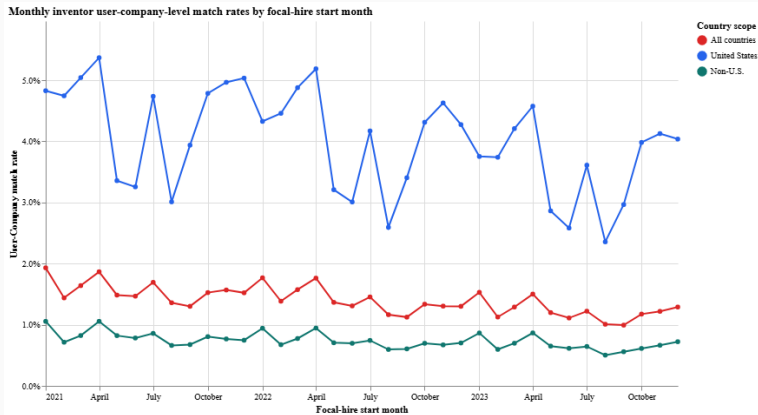
- US is not the country with the highest match rates among the candidate new hires.
- Top 5 economies: South Korea (9.12%); Japan (8.13%); Taiwan, China (5.69%); US (3.89%); Israel (3.85%).

Match rates are significantly larger for high-seniority new hires



- New hires who are more senior are more likely to show up in the inventor database.
- This is intuitive but has important implications for our sample construction:
 - Are we measuring productivity or seniority among the new hires?
 - A company who hires more senior new hires will have higher observed productivity based on patents. However, it is not necessarily more productive.

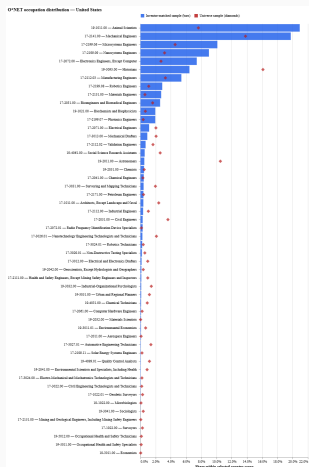
Time series for match rates (start time)



Outline

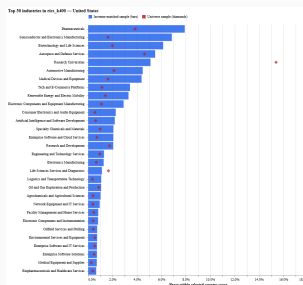
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 - US matched sample
 - All-country matched sample
 - Other results

ONET occupation distribution in the matched sample (US)



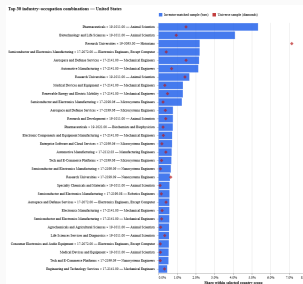
- Top 5 ONET occupations in the matched sample within US new hires: Animal Scientists (20.9%); Mechanical Engineers (19.7%); Microsystems Engineers (10.1%); Nanosystems Engineers (9.0%); Electronics Engineers, Except Computer (7.4%).
- The top 5 ONET occupations constitute around 70% of the matched sample.

Industry distribution in the matched sample (US)



- Top 5 Revelio 400-category industries: Pharmaceuticals (7.9%); Semiconductor and Electronics Manufacturing (6.9%); Biotechnology and Life Sciences (6.1%); Aerospace and Defense Services (5.5%); Research Universities (5.1%). The matched sample is therefore much more concentrated in patent-intensive industries than the universe sample.

Industry-ONET distribution in the matched sample (US)

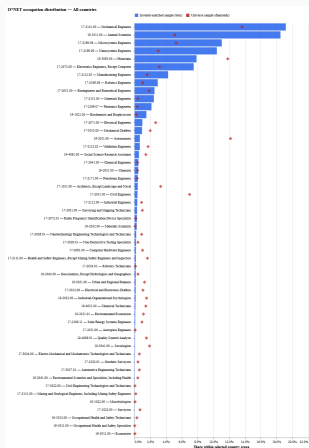


- Top 5 industry–occupation cells: Pharmaceuticals × Animal Scientists (5.3%); Biotechnology and Life Sciences × Animal Scientists (4.1%); Research Universities × Historians (2.2%); Semiconductor and Electronics Manufacturing × Electronics Engineers, Except Computer (2.2%); Aerospace and Defense Services × Mechanical Engineers (2.2%). These cells illustrate that inventor linkage changes the joint industry–occupation composition, not only each marginal distribution.

Outline

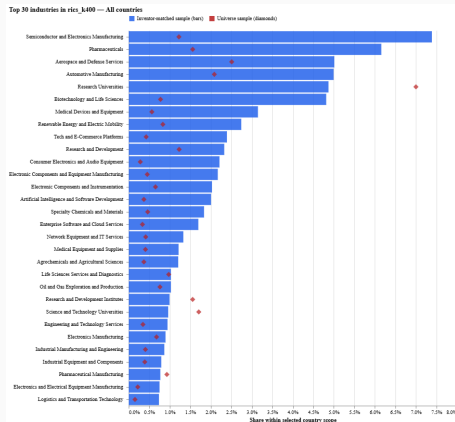
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ONET occupation distribution in the matched sample (all countries)



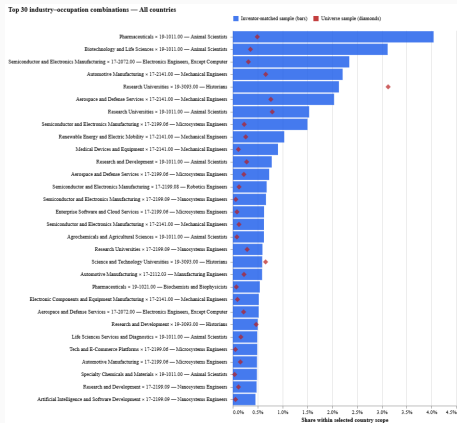
- Top 5 ONET occupations in the matched sample within all-country new hires: Mechanical Engineers (19.2%); Animal Scientists (18.5%); Microsystems Engineers (11.1%); Nanosystems Engineers (10.4%); Historians (7.9%).
- Again, the high share of “Historians” in the matched sample indicates how messy Revelio’s occupation classifications are.

Industry distribution in the matched sample (all countries)



- Top 5 Revelio 400-category industries: Semiconductor and Electronics Manufacturing (7.4%); Pharmaceuticals (6.2%); Aerospace and Defense Services (5.0%); Automotive Manufacturing (5.0%); Research Universities (4.9%). Relative to the universe, the matched sample shifts toward industries in which patenting and inventor linkage are more prevalent.

Industry-ONET distribution in the matched sample (all countries)

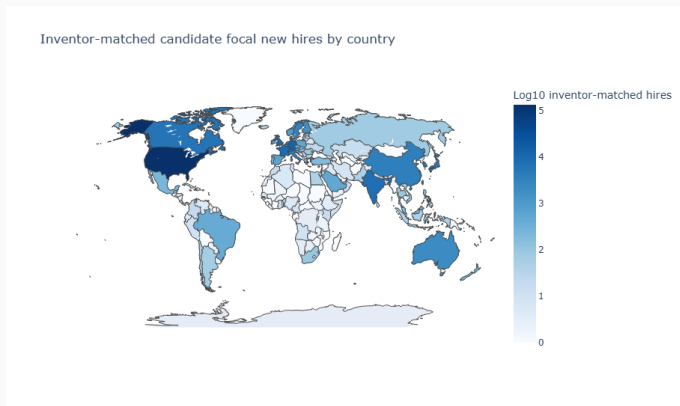


- Top 5 industry-occupation cells: Pharmaceuticals × Animal Scientists (4.0%); Biotechnology and Life Sciences × Animal Scientists (3.1%); Semiconductor and Electronics Manufacturing × Electronics Engineers, Except Computer (2.3%); Automotive Manufacturing × Mechanical Engineers (2.2%); Research Universities × Historians (2.1%).

Outline

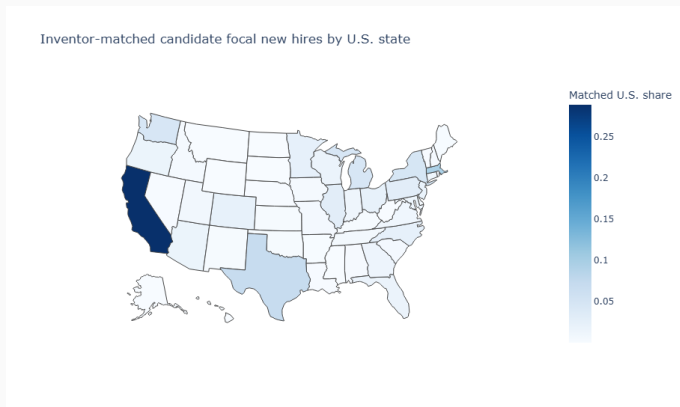
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Matched sample across countries



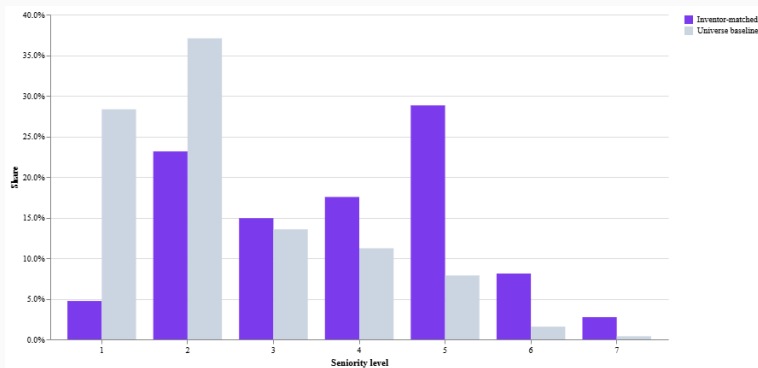
- Top 5 countries in the matched sample: US (58.5%); Germany (5.9%); France (4.0%); UK (3.8%); India (3.3%). The US share rises from 21.3% in the universe to 58.5% in the matched sample, so inventor linkage substantially changes the geographic composition.

Matched sample across states in US



- Top 5 states in the matched US sample: California (28.9%); Massachusetts (9.7%); Texas (7.1%); New York (4.9%); Michigan (4.7%). California's share rises from 16.7% in the US universe to 28.9% in the matched sample.

Matched sample has a different seniority distribution compared with all candidate new hires



- Again, as I have said before, this could have serious implications for our interpretations of the patents-based productivity results.

Monthly count of inventor-matched candidate focal new hires

